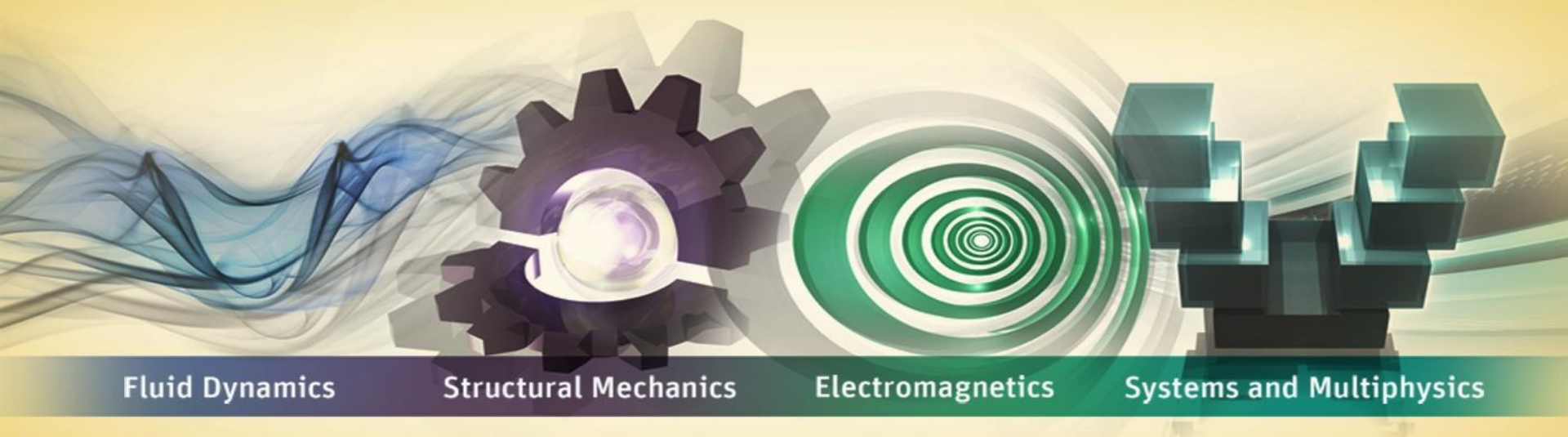




Lecture 0: GUI overview

Realize Your Product Promise™

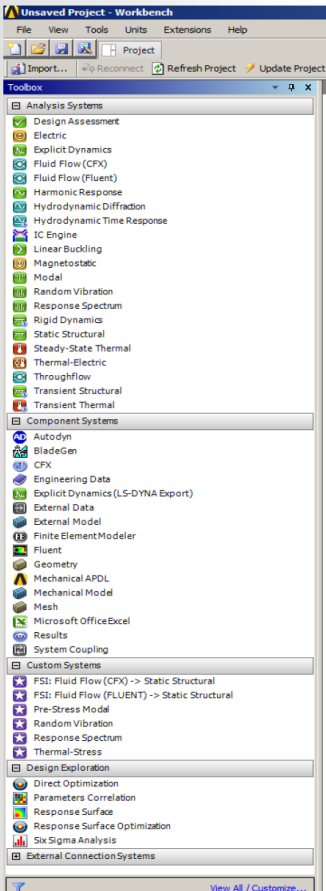


ANSYS WORKBENCH

The options visible in the left-hand column show all the products (systems) you have licenses for.

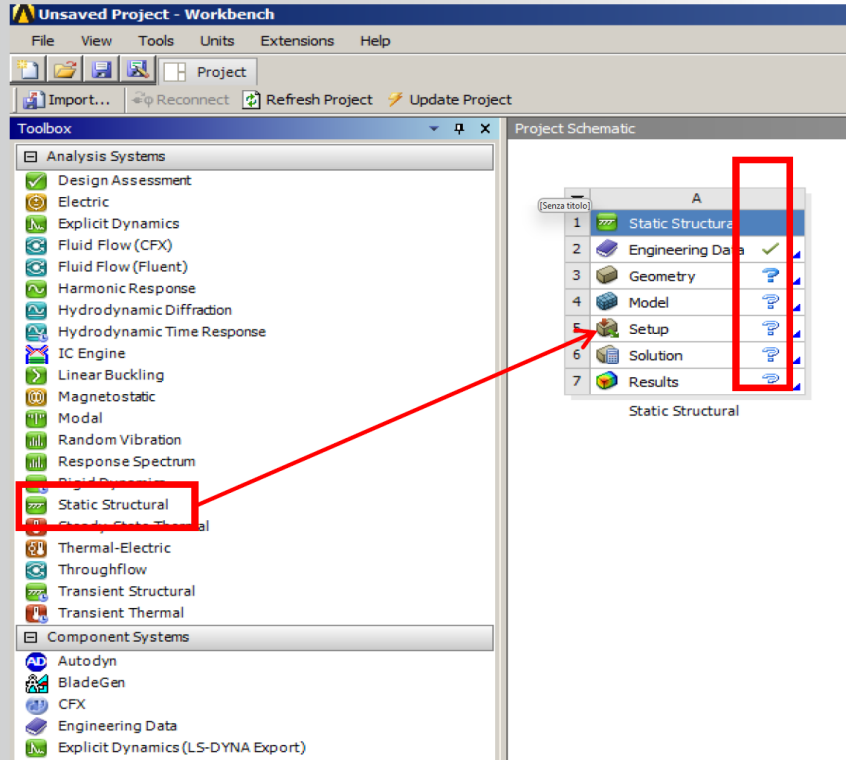
TIP: If this list appears empty, you have a problem with your licensing.

“Design Exploration” provides tools for optimising designs and understanding the parametric response.



“Analysis Systems” are ready-made stencils that include all the individual systems (applications) needed for common analyses (for example Geometry + Mesh + Solver + Post-Processor)

“Component Systems” are the individual building-blocks for each stage of the analysis



Dragging an **Analysis System** onto the project desktop lays out a workflow, comprising all the steps needed for a typical analysis.

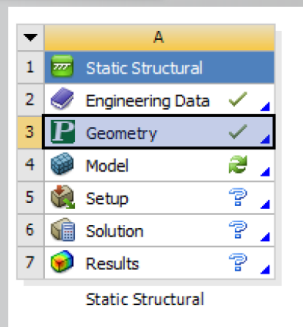
Workflow is from top to bottom. As each stage is complete, the icon at the right-hand side changes

The screenshot shows the ANSYS Workbench interface. On the left is the 'Toolbox' with 'Analysis Systems' and 'Component Systems'. In the center is the 'Project Schematic' showing three cells: A (Geometry), B (Mechanical Model), and C (Static Structural). Blue lines represent connectors between cells. A red arrow points from the 'Static Structural' icon in the toolbox to the 'Static Structural' cell in the schematic.

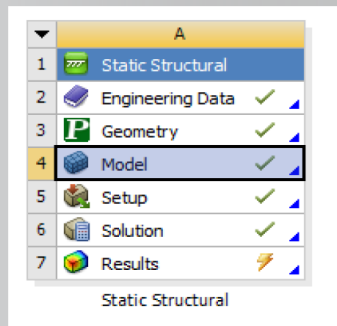
However, an analysis could equally well be prepared by selecting the individual **Component Systems** that are needed for this analysis, and then linking them together with connectors.

TIP: There are two ways to create the connectors between the systems:

- 1) Use the mouse to draw a line (eg A2 to B3, B4 to C4 etc)
- 2) Or, simply drop the new system on the cell of the upstream one, and the link will be generated automatically.



**Status after creating Geometry in A3,
not yet opened Mechanical in A4**



**Status after model has solved
waiting for post-processing**

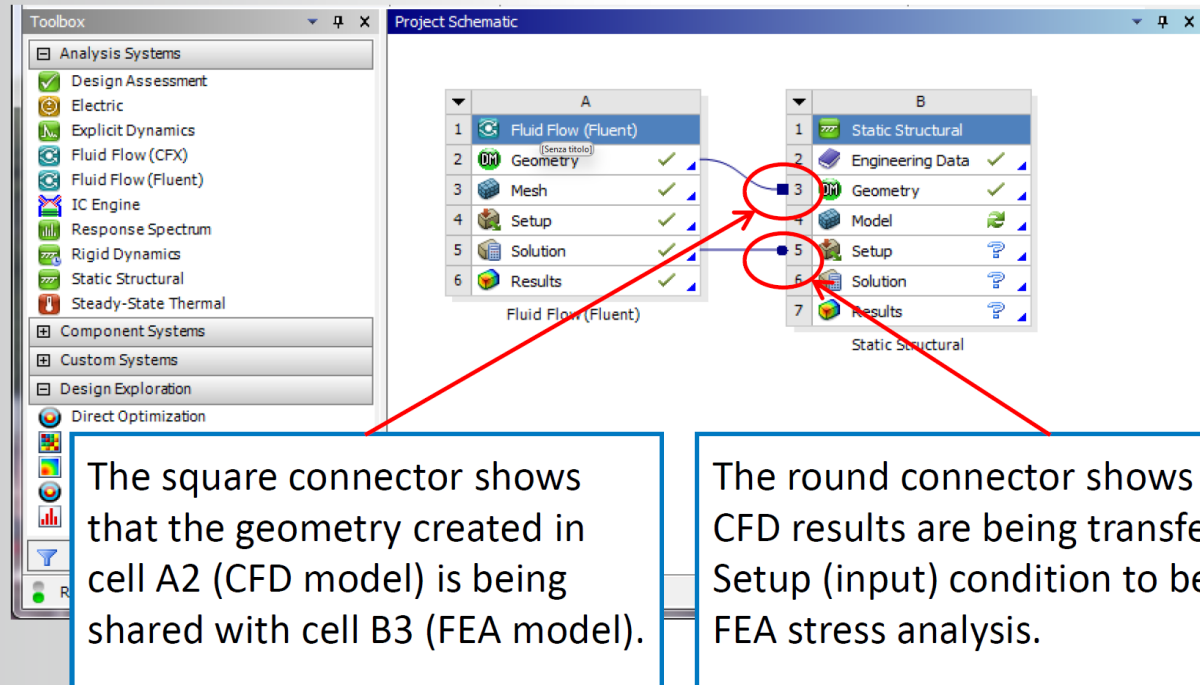
As each stage in the model-build is completed, the state of the cell changes.

Icon Meaning

	Up to Date
	Refresh required. Upstream data has changed
	Update required. Local data has changed
	Unfulfilled. Upstream data does not exist
	Attention Required
	Solving
	Update Failed
	Update Interrupted
	Changes pending (was up-to-date, but upstream data has changed)

... Sharing Data between Different Solvers

Workbench can be used to transfer data between solvers. In this 1-way FSI (fluid-structure-interaction) example, we transfer the loads from a Fluent CFD simulation over to a Mechanical system to perform a stress analysis



... File Location on Disk

Should you need to identify the individual files on your disk for each stage of the project, these can be found by enabling View > Files. The resulting table will cross-reference the directory and filename with the project cells.

The screenshot shows the ANSYS Workbench interface. The 'View' menu is open, and the 'Files' option is highlighted with a red circle. A red arrow points from this menu item to the 'Files' table in the lower right. The 'Files' table lists project files with columns for Name, Cell ID, Size, Type, Date Modified, and Location. A red box highlights the 'training.inpq' and 'SYS.mechdb' files in the 'Name' column. Another red box highlights the 'Location' column for these files, showing paths like 'D:\Training\15.0\training_files\vp0\global\MECH'.

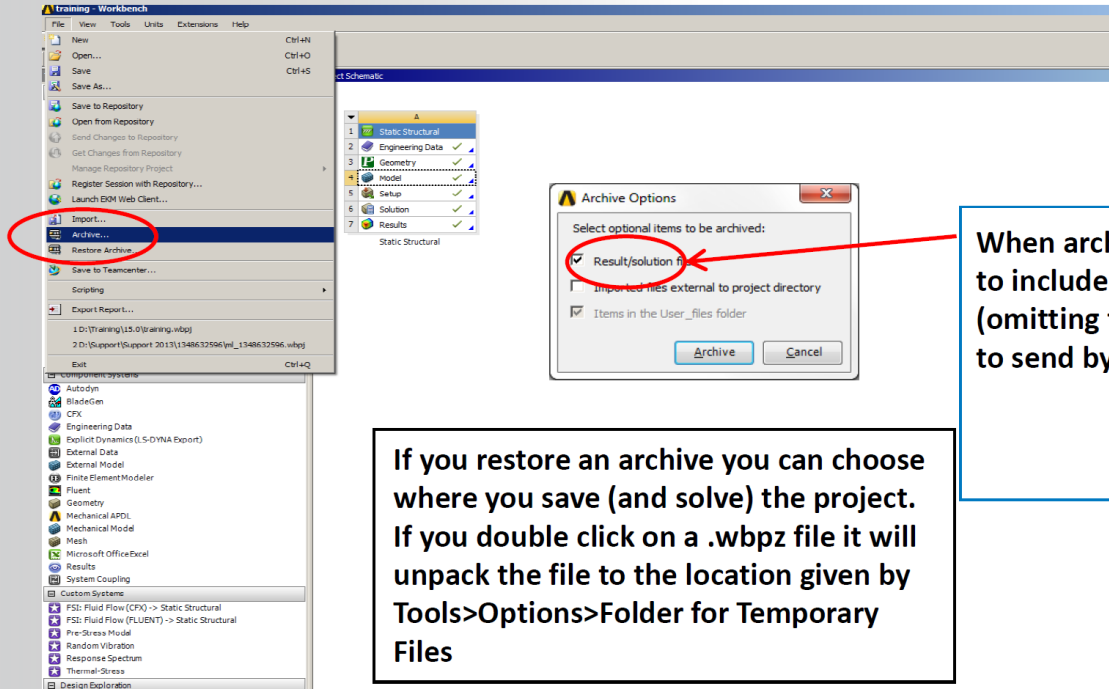
	A	C	D	E	F	
	Name	Cell ID	Size	Type	Date Modified	Location
1						
2	IntroDemo.x_t	A3	58 KB	Geometry File	01/06/1999 19:42:32	C:\Users\mledin\Documents\Sample Geometry Files
3	material.engd	A2	18 KB	Engineering Data File	03/12/2013 11:21:27	D:\Training\15.0\training_files\vp0\sys\mech
4	inprt.inpq	A4	18 KB	Engineering Data File	03/12/2013 11:21:27	D:\Training\15.0\training_files\vp0\global\MECH
5	training.inpq	A4	141 KB	Workbench Project File	03/12/2013 11:22:20	D:\Training\15.0
6	SYS.mechdb	A4	124 KB	Mechanical Database File	03/12/2013 11:22:20	
		A2	17 KB	Engineering Data File	03/12/2013 11:22:20	
		A1	12 KB	.xml	03/12/2013 11:21:27	
		A1	849 B	.xml	03/12/2013 11:22:10	
		A1	229 KB	.det	03/12/2013 11:21:27	D:\Training\15.0\training_files\vp0\sys\mech
		A1	2 KB	.jcs	03/12/2013 11:22:09	D:\Training\15.0\training_files\vp0\sys\mech

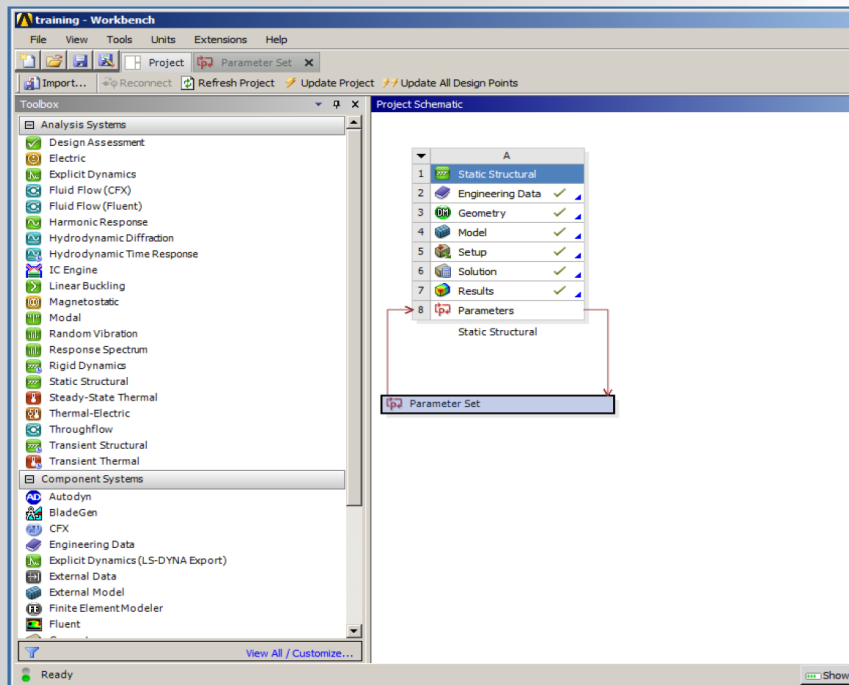
Filename

Directory

... Use of Archive / Restore

The workbench project comprises many files and directories. If you need to either archive the project, or bundle it to send to us for a Technical Support query, use the '**Archive**' tool. This generates a single zipfile of the entire project.





Most Workbench applications will let you specify key quantities as a **parameter** (rather than a constant). This will be covered later.

In this example:

- When creating the geometry in DesignModeler, hole diameter is set to be an *input parameter*.
- When reviewing the results, the maximum stress is set as an *output parameter*

We could just have easily set up a CFD analysis, looking at different loading conditions and reporting the pressure drops.

The screenshot shows the ANSYS Workbench Parameter Set interface. The 'Outline of All Parameters' table on the left lists input and output parameters. The 'Table of Design Points' table on the right shows the results of four design points. Four numbered callouts provide instructions:

- 1] Create new row in table with each design (in this case hole diameter)
- 2] Click here to compute all the designs
- 3] The desired result (set up in Mechanical) is reported here
- 4] The 'Exported' option allows snapshots of each DP to be saved to a different project

Table of Design Points

	A	B	C	D
1	Name	P1 - hole_dia	P3 - Equivalent Stress 2 Maximum	Exported
2	Units	m	MPa	
3	Current	0.05	32.272	
4	DP 1	0.055	35.469	☐
5	DP 2	0.06	39.992	☐
=				

- Clicking on 'Parameter Set' lets us vary these parameters.
- Four different geometric designs are being tested.
- The whole process is automated. Workbench will recursively:
 - Create the geometry, based on the parameters in the table
 - Take this into Mechanical and remesh and solve and then the postprocessor
- The user just needs to sit back and wait, and the matrix of experiments (each requiring several different applications to be launched in turn) is computed automatically.